Thomas C. Fraser

tfraser@perimeterinstitute.ca
+1 (226) 868-0557
31 Caroline St N, Office 462
Waterloo, ON N2L 2Y5, Canada
November 2022

Education

2018 – Present	Ph.D., Quantum Information *Perimeter Institute For Theoretical Physics, Waterloo, ON*
2017 – 2018	M.Sc., Theoretical Physics, Quantum Information Thesis: A Combinatorial Solution to Causal Compatibility *Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON*
2012 – 2017	B.Sc., Mathematical Physics, Astrophysics Specialization University of Waterloo, Waterloo, ON

Papers

2022 T. C. Fraser
A sufficient family of necessary inequalities for the compatibility of quantum marginals
arXiv preprint arXiv:2211.00685

2020 D. Schmid, T. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens
Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory
arXiv preprint arXiv:2004.09194

2020 T. C. Fraser
A Combinatorial Solution to Causal Compatibility
Journal of Causal Inference 8(1), 22-53

2018 J. R. Rinehart, J. H. Béjanin, T. C. Fraser, M. Mariantoni
Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry
arXiv preprint arXiv:1804.04756

2018 T. C. Fraser, E. Wolfe
Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario
Physical Review A, 98(2), 022113

Talks

2022 The IQC-QuICS Math and Computer Science Seminar
A sufficient family of necessary inequalities for the quantum marginals problem

2021 Beyond IID in Information Theory 9
Simultaneous Spectral Estimation and the Quantum Marginal Problem

2021 PI Grad Student Conference, Waterloo, Canada
Simultaneous Spectral Estimation and the Quantum Marginal Problem

2019 IQST Seminar at University of Calgary, Canada
Sudoku on steroids - The combinatorics of causal inference

2016 Quantum Networks Conference at International Institute for Physics, Natal, Brazil
Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario

Teaching

Winter 2021	Quantum Information - Perimeter Scholars International (PSI) Program *Teaching Assistant*
Winter 2021	Quantum-classical gaps in causal structures - PSI Winter School *Student project co-lead*
Fall 2018	Physics 1 - Phys 111 - University of Waterloo *Teaching Assistant*

Awards & Scholarships

2022	Natural Sciences and Engineering Research Councilof Canada (NSERC) *Three years, Grant No. 411301803*
2018	UW Graduate Award & Marie Curie Award *Sufficiently High Entrance Average, Department Selected*
2018	The Joanne Cuthbertson and Charlie Fischer Graduate Student Award *Support for an exceptional Masters student from the 2017/2018 PSI class*
2017	Governor General's Silver Academic Medallion *Highest Graduating Average in Graduating Class*
2016	Mike Lazaridis Scholarship *Theoretical Physics Fellowship at Perimeter Institute*
2015	Xerox Research Centre of Canada Limited Award *Best work-term report "Acoustic Modelling Using Mel-Frequency Cepstral Coefficients"*
2015	C. C. Lim Physics Prize *Top Marks in Undergraduate Thermodynamics*
2013	Don E. Brodie Scholarship *Highest Experimental Physics Lab Performance*
2012	A. Donald Maynes Scholarship *Outstanding Academic Record*
2012	BMO Undergraduate Entrance Scholarship *Outstanding Academic Average*

Industry Experience

September 2015 – January 2016	Research & Development Data Scientist Sysomos, Toronto, ON, Canada
September 2014 – May 2015	Video Game Developer Lunarch Studios, Waterloo, ON, Canada
January 2014 – May 2014	Mathematics Tutor Humber College, Toronto, ON, Canada
June 2011 – August 2011	Solar Panel Technician OVG Solar, Inc, Renfew, ON, Canada

Programming

Languages	C, C++, Python, Mathematica, Matlab, Actionscript, JavaScript